

Cloud Computing Services

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Infrastructure as a Service (IaaS)

- 1 Contains The Basic Building Blocks
 - Virtualization
- 2
 - Servers
 - Storage
 - Networking
- 3 Example
 - AWS EC2
 - Azure VM
- 4
 - Advantage
 - The most flexible and dynamic model
 - Cost-effective due to pay-as-you-go pricing
 - Easy to use due to the automate d deployment of hardware
 - Management tasks are virtualized, so employees have more free time for other tasks
 - Disadvantage
 - Data security issues due to multitenant architecture
 - Vendor outages make customers unable to access their data for a while
 - The need for team training to learn how to manage new infrastructure
- 5 Use Case
 - If you are a small company or a startup that has no budget for creating your own infrastructure
 - If you are a rapidly growing company and your demands are unstable and changeable
 - If you are a large company that wants to have effective control over infrastructure but pay only for the resources you actually use

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Platform as a Service (PaaS)

- 1 The Underlying Infrastructure (usually hardware and operating systems)
- 2
 - O/S
 - Middleware
 - Runtime
- 3 Example
 - AWS Elastic Beanstalk
 - AWS RDS
 - AWS S3
- 4
 - Advatage
 - 1 PaaS-built software is highly scalable, available and multi-tenant, as it is cloud-based
 - 2 The development process is quickened and simplified
 - 3 Reduced expenses for creating, testing and launching apps
 - 4 Automated company policy
 - 5 Reduced amount of coding required
 - 6 Allows for easy migrating to the hybrid cloud
 - Disadvantage
 - 1 Data security issues
 - 2 Compatibility of existing infrastructure (not every element can be cloud-enabled)
 - 3 Dependency on vendor's speed, reliability and support
- 5 Use Case
 - 1 Multiple developers work on one project
 - 2 Other vendors must be included
 - 3 You want to create your own customized apps

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Software as a Service (SaaS)

- 1 Completed product that is run and managed by the service provider.
- 2
 - Data
 - Application
- 3 Example
 - Google Apps
 - Office 365
- 4
 - Advantage
 - 1 No hardware costs
 - 2 No initial setup costs
 - 3 Automated upgrades
 - 4 Cross-device compatibility
 - 5 Accessible from any location
 - 6 Pay-as-you-go model
 - 7 Scalability
 - 8 Easy customization
 - Disadvantage
 - 1 Loss of control
 - 2 Limited range of solutions
 - 3 Connectivity is a must
- 5 Use Case
 - 1 If your company needs to launch a ready-made software quickly
 - 2 For short-term projects that require collaboration
 - 3 If you use applications on a temporary basis
 - 4 For applications that need both web and mobile access